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REVIEW ARTICLE

Overlapping Mechanisms of Exertional Heat Stroke and Malignant Hyperthermia: Evidence vs. Conjecture

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Abstract

Exertional heat stroke (EHS) and malignant hyperthermia (MH) are life-threatening conditions, triggered by different environmental stimuli that share several clinical symptoms and pathophysiological features. EHS manifests during physical activity normally, but not always, in hot and humid environments. MH manifests during exposure to haloalkane anesthetics or succinylcholine, which leads to a rapid, unregulated release of calcium (Ca^{2+}) within the skeletal muscles inducing a positive-feedback loop within the excitation–contraction coupling mechanism that culminates in heat stroke-like symptoms, if not rapidly recognized and treated. Rare cases of awake MH, independent of anesthesia exposure, occur during exercise and heat stress. It has been suggested that EHS and MH are mediated by similar mechanisms, including mutations in Ca^{2+} regulatory channels within the skeletal muscle. Rapid cooling, which is the most effective treatment for EHS, is ineffective as an MH treatment; rather, a ryanodine receptor antagonist drug, dantrolene sodium (DS), is administered to the victim to prevent further muscle contractions and hyperthermia. Whether DS can be an effective treatment for EHS victims remains uncertain. In the last decade, multiple reports have suggested a number of mechanistic links between EHS and MH. Here, we discuss aspects related to the pathophysiology, incidence, diagnosis and treatment. Furthermore, we present evidence regarding potential overlapping mechanisms between EHS and MH and explore current knowledge to establish what is supported by evidence or a lack thereof (i.e. conjecture).

Key Points

- Exertional heat stroke (EHS) and malignant hyperthermia (MH) are believed to share mechanistic links.
- Calcium dysregulation in skeletal muscle is proposed to be one of the main mechanistic links between EHS and MH.
- Whole body cooling is effective to treat EHS, but not effective to treat MH.
- Dantrolene sodium is effective to treat MH, but not clearly effective to treat EHS.

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1 Introduction

Exertional heat stroke (EHS) and malignant hyperthermia (MH; also known as malignant hyperpyrexia, hyperthermia of anesthesia, ORPHA423) are hypermetabolic states that can result in death if inappropriately managed [1, 2]. EHS, the most severe manifestation of heat-related illnesses, is a result of cardiovascular strain due to competing demands for blood between peripheral tissues for heat dissipation and skeletal muscles for contractions during physical exertion [2, 3]. Although EHS may also occur in the absence of environmental stress [4], it is often exacerbated in hot and humid environments [3]. MH is a pharmacogenetic disorder that manifests during exposure to succinylcholine or the haloalkane class of anesthetic gases such as halothane, sevoflurane, desflurane, isoflurane. Exposure to these triggering agents stimulates a massive calcium (Ca^{2+}) release within skeletal muscles resulting in sustained contractions leading to severe hyperthermia [1, 5, 6]. Awake cases of MH have also been reported in response to vigorous exercise and heat stress (i.e. in the

absence of anesthesia exposure) and, therefore, awake MH cases can be difficult to distinguish from EHS [7–14].

Exertional heat stroke (EHS) and MH can occur in young, otherwise healthy individuals and both are associated with a high metabolic (ATP) demand, accelerated oxidative, chemical and mechanical stress of skeletal muscles and presumably uncontrolled increases in intracellular Ca^{2+} [13, 15–17]. Although controversial [18], it has been suggested that EHS may be a consequence of Ca^{2+} dysregulation in the skeletal muscle that mimics that observed with MH [17, 19]. As such, Ca^{2+} dysregulation, which is an underlying feature of MH, is frequently listed as a coexisting feature in both conditions. However, as will be discussed throughout this review, it is unclear whether Ca^{2+} dysregulation is involved in EHS pathophysiology.

Our goal in this review is to discuss these and other aspects of EHS and MH to delineate similarities and differences between these conditions regarding pathophysiology, incidence, diagnosis and treatment, and to identify knowledge gaps that need to be addressed to more accurately determine potential overlapping mechanisms that mediate the two conditions.

2 Pathophysiology of Exertional Heat Stroke and Malignant Hyperthermia

The presence of hyperthermia is obviously the first overlapping pathophysiological characteristic of EHS and MH. One of the main features of EHS is the inability to rapidly dissipate metabolic heat generated by skeletal muscle contractions during exertion [2, 3, 20]. Skeletal muscle contractions are only ~ 20% efficient (i.e. energy used to resynthesize ATP from ADP) such that ~ 80% of the energy is released in the body as heat, which must be dissipated to avoid hyperthermia. Under compensable environmental conditions, heat can be effectively dissipated to the environment, mainly through sweat evaporation, so that the rate of heat loss remains in balance with heat production and a steady-state body core temperature is maintained [20]. However, under uncompensable conditions (e.g., high environmental temperature; protective clothing ensembles; high relative humidity) that compromise the dissipation of heat to the environment, a high cardiac output cannot be sustained and severe hyperthermia ensues leading to collapse [3, 21]. The EHS victim will exhibit central nervous system (CNS) dysfunction (delirium, combativeness, seizures, coma) with hyperthermia that can progress to multi-organ damage and failure if not properly treated [18]. Upon collapse, the victim typically exhibits muscle flaccidity, which may be suggestive of absence of dysfunctional Ca^{2+} homeostasis, that halts metabolic heat production and aids in passive cooling [2]. EHS remains a more preventable than treatable condition because of limitations in our understanding of the mechanisms mediating organ injury. However, rapid effective cooling using techniques such as ice water immersion or iced sheets will accelerate the rate of cooling and prevent injury, under most conditions [22–24]. There are currently no pharmacological interventions to treat EHS, although this is currently an active area of research.

In contrast to EHS, the distinguishing clinical feature of MH is intense whole-body sustained spasmodic skeletal muscle contractions, which result in substantial metabolic heat generation leading to heat stroke, when untreated [25]. There are ~ 600 skeletal muscles in the human body and the simultaneous sustained contraction of a large subset of muscles, which occurs in anesthesia-induced MH, can result in lethal hyperthermia. The increased metabolism with MH is associated not only with muscle rigidity, but also with cyanosis, hyperkalemia, myoglobinuria, tachycardia, cardiac arrhythmia, unstable arterial pressure, severe metabolic and respiratory acidosis [26] with later complications that may include renal failure, consumptive coagulation, and hypokalemia [27]. It is well accepted that MH is a consequence of uninterrupted activation of a skeletal muscle Ca^{2+} release positive feedback loop that can only be treated with pharmacological intervention, specifically dantrolene sodium (DS), a ryanodine receptor antagonist, that disrupts that cycle [5].

Because MH culminates in EHS-like symptoms due to the excessive metabolic heat generated during the sustained skeletal muscle contractions (Fig. 1), it is no surprise that these two conditions share similar clinical symptoms [15, 17, 19]. Nonetheless, EHS is a consequence of combined environmental and metabolic heat generation from only a few, albeit large, active skeletal muscle groups that are directly involved in the physical activity [18, 28]. Perhaps more importantly, EHS patients exhibit muscle flaccidity, rather than rigidity, upon collapse which halts metabolic heat production and aids in cooling. This behavioral distinction between EHS and MH is the first evidence that suggests Ca^{2+} dysregulation may not be a mechanistic link between these conditions, as has been hypothesized in recent years [17, 19]. Besides the rigidity versus flaccidity differences, there are notable contrasts regarding the incidence of these two manifestations.

3 Incidence of Exertional Heat Stroke and Malignant Hyperthermia

Heat stroke manifests in two distinct forms that differ in incidence. Classic (i.e. passive) heat stroke (CHS) is the most common form that is observed in the very young or elderly during annual heat waves. The incidence of CHS

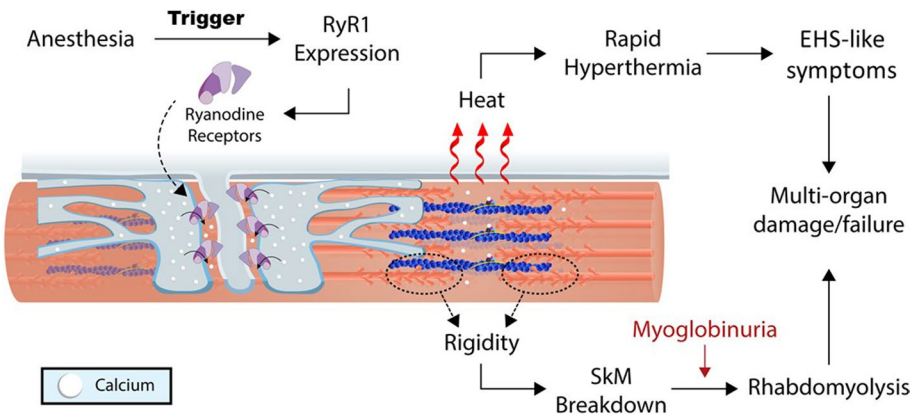

Fig. 1 Hypothesized overlapping mechanisms observed in awake cases of MH leading to EHS-like symptoms. Exposure to haloalkane anesthetics triggers Ca^{2+} release within the myofibers. Sustained contractions and heat production cause hyperthermia. Skeletal muscle rigidity may lead to rhabdomyolysis (i.e. skeletal muscle breakdown) with myoglobin leakage, which contributes to multi-organ injury. *RyR1* ryanodine receptor 1, *SkM* skeletal muscle

typically ranges from ~ 200 to 300 cases per year (although larger incidence has occurred more recently) with the severity often exacerbated by pre-existing conditions, such as cardiovascular insufficiency, medications or illness [2, 29]. EHS occurs during physical activity in hot or temperate environments and often under conditions previously well tolerated without collapse or sequelae [4, 30]. Furthermore, EHS is most often observed in young, otherwise healthy individuals that do not have pre-existing medical conditions. Despite being the third leading cause of mortality in athletes during physical activity [31], EHS incidence tends to be much lower than CHS, presumably due to treatment at the site of collapse that evades clinical reporting (e.g., marathon treatment tents), misdiagnosis due to signs and symptoms that overlap with other exercise-induced conditions (e.g., hyponatremia) [2], or body temperature recordings at peripheral sites (e.g., axilla, tympanic membrane, forehead) or after cooling that results in underestimation of core temperature and complicates accurate clinical diagnosis. Recently, Knight et al. [32] retrospectively determined EHS incidence for 11 consecutive years of triathlons using diagnostic criteria that included body core temperature > 40 °C with evidence of an altered mental state indicative of neurological dysfunction. Out of 68,557 starting competitors, only 12 were diagnosed with EHS, for an incidence of 1.74/10,000 starters. Despite differences in etiology, rapid, effective cooling remains the most effective treatment for both CHS and EHS to mitigate organ injury and ensure low mortality rates [32]. For a more in-depth discussion on whole body cooling in EHS, readers are referred to Laitano et al. [18].

The incidence of MH is not known with certainty, but estimates suggest that general anesthesia triggers the response in a range as wide as 1:5000 to 1:50,000–100,000 individuals [33]. These numbers are possibly underestimated due to some individuals never being exposed to a triggering agent, the duration of exposure being too low or too brief to trigger a clinical response, or it may require multiple exposures before a triggering event occurs. Indeed, a patient requires, on average, three anesthetic exposures before the clinical signs manifest [34], even though there are reports of episodes occurring during the first exposure. It is likely that the etiology is multi-factorial with prior illness, stress or other events causing the trigger threshold to decrease with each subsequent exposure (Table 1). Results of recent molecular studies estimate that the genetic prevalence, which corresponds to the number of people carrying the mutation, may be as high as 1:2000–3000 due to the autosomal dominant inheritance of the trait [35–37]. MH can develop in children as early as 6 months of age with prevalence that is 2.5–4.5 times higher in males than females [38]. Sex differences may be explained by the fact that females have a smaller muscular build and lower Ca^{2+}-ATPase activity than males [39]. In fact, larger muscular build, which is most common in males, is considered a potential pre-disposing factor for MH [16]. For individuals who have a sudden onset of MH symptoms, quick action taken by medical professionals will subdue the clinical progression to prevent long-term organ injury or death. Unfortunately, rapid cooling with ice packs or other external measures is not sufficient to prevent heat stroke symptoms in MH patients [40]; rather, DS, a specific ryanodine receptor antagonist, is the only established MH clinical treatment that is effective when administered early during the progression of an episode [41]. DS targets the ryanodine 1 receptor (RyR1) in skeletal muscle, which was originally considered the only defective receptor in MH patients—other defective receptors and mutations associated with MH continue to be identified

Table 1 Studies reporting awake MH with either exercise or heat exposure

Case presentation	Risk factors	Treatment	Diagnosis	Comments	References
Family (630 persons) with high awake MH incidence	None identified	None	Stress syndrome	Similarities to the porcine stress syndrome; obvious autosomal dominant inheritance	[7]
15 yo Caucasian male found rigid on ground after jogging	URI 1 week prior	None	**Physical Examination (recent)** Normal **Gross Post Mortem Examination** Hilar lymphadenopathy Cardiomegaly SkM normal histology	Father MH positive (Ca^{2+} uptake test) Cardiomyopathy common with MH	[70]
20 yo African American male with frequent muscle cramps during training	Muscular build (6′2″, 185 lb)	IV fluids Cooling	**NAMHG protocol** ✓ halothane o caffeine *Diagnosis: MH susceptible*	Tc rose AFTER ceasing exercising (does not normally happen with EHS); suggests a link between MH and EHS, but not proven	[69]
6 yo Caucasian male hyperthermic (108.9°F), sweating profusely with muscle rigidity after playing in hot, humid environment	Lumbar lordosis	IV fluids and external ice packs ineffective	**Genetic analysis** RyR1 variant *Diagnosis: awake MH reaction*	Father and 2 siblings with lumbar lordosis had same RyR1 variant Father positive for halothane and caffeine; diagnosed with central core disease	[8]
120 MH patients	Family history	Not stated	**NAMHG protocol** 21% MHS 7% MHN 73% MHN **Morphological analysis** Normal histology	Despite anesthesia or other MH episodes, IVCT only detected 21% of cases	[9]
45 EHS cases	None	Not stated	**NAMHG protocol** 24% MHS 15% MHE 58% MHN **Morphological analysis** Normal histology	IVCT never validated for EHS	
42 yo male with intermittent fever of unknown origin	Muscular build Slightly obese	Aspirin and cooling ineffective Sensitive to oral dantrolene	**EMHG protocol** o halothane ✓ caffeine *Diagnosis: MH susceptible due to hypermetabolic stress response*	Father muscular build with high CPK values Father normal anesthesia exposures Daughter and son normal CPK values, normal IVCT, normal anesthesia exposures	[11]

Table 1 (continued)

Case presentation	Risk factors	Treatment	Diagnosis	Comments	References
12 yo male with jaw clenching, hyperthermia (> 42.2 °C), seizures after football practice	Ventricular hypertrophy	Dantrolene	**SkM histology** Normal **DNA analysis** RyR1 variant *Diagnosis: non-anesthetic stress-induced hyperpyrexic death with history of MH*	Same RyR1 variant in father Observers denied exertion or excessive environmental conditions (80°F)	[1]
3 yo male with recurrent awake MH episodes	Central core disease (ptosis; hypotonia) Mild upper respiratory/GI infection	Aggressive cooling Antipyretics (paracetamol) Dantrolene	**Electromyography** Congenital myopathic discharge pattern **NAMHG protocol** ✓ Halothane ✓ Caffeine **Genomic DNA analysis (B lymphocytes)** RyR1 heterozygous nucleotide change (R3983C) *Diagnosis: atypical carnitine palmitoyl-transferase 3 deficiency; de novo R3983C variant*	Both parents lacked the R3093C RyR1 variant	[73]
6 yo girl with fever and rigidity in > 1 incident	Recent viral illnesses in both instances	Dantrolene ineffective Ice bath, oral antipyretics and IV fluids effective	**Genomic DNA analysis (B lymphocytes)** RyR1 heterozygous nucleotide change (R3983C) *Diagnosis: de novo R3983C variant*	No MH family history	
19 yo soldier collapsed with EHS (41 °C) during march run in combat clothing	None	Sponging IV fluids	**EMHG protocol** ✓ Halothane o Caffeine o Ryanodine *Diagnosis: EHS associated with underlying inherited SkM abnormality*	Mother IVCT normal Father positive to halothane, negative to caffeine and ryanodine	[14]
23 yo soldier with recurrent EHS (axillary temperature = 39 °C)			**EMHG protocol** ✓ Halothane o Caffeine – Abnormal ryanodine *Diagnosis: MH equivocal*	Mother IVCT normal Father IVCT normal with abnormal ryanodine Brother IVCT normal with abnormal ryanodine EHS SkM Ca^{2+} dysregulation that is distinct from MH	

Table 1 (continued)

Case presentation	Risk factors	Treatment	Diagnosis	Comments	References
30 yo male semi-pro rugby player seized during a race	None	Rotated ice bags Plasma Intensive hydration	**Targeted mutational analysis** ✓ RyR1 heterozygous missense mutation (p.K1393R) *Diagnosis: EHS with ER associated with MH susceptibility*	No MH episodes or other abnormal responses documented in this individual Strong diagnosis given limited evidence	[13]
466 EHS patients in the French Armed Forces (retrospective)	Prior anesthesia in 50% of patients Prior EHS (12.7%) Strenuous, prolonged exercise	Unknown	**EMHG protocol** ✓ Halothane (18.5%) ✓ Caffeine (9.9%) ✓ Both (17.2%) *Diagnosis: latent disturbance of Ca^{2+} homeostasis in EHS patients*	EHS definition; mental status changes with Tc > 39 °C (rectal or aural) RyR1 sequence variants: MHN > MHS patients Dantrolene never administered Need to separate out those with prior EHS or prior anesthesia EHS recurrence not associated with MH status	[19]
16 yo male with EHS and ER after football practice with altered mental status, hyperthermia (42.2 °C)	Hospitalization for ER 7 days prior Early in training season Muscular bulk (135 kg)	Dantrolene with cooling	**NAMHG protocol** o Halothane o Caffeine *Diagnosis: EHS with no MH susceptibility*	Dantrolene has been shown to be effective with ER, but not EHS; may explain efficacy	[12]
17 yo male with post-exercise muscle cramping in hot, humid weather	None identified	Limit exercise in hot, humid weather Oral dantrolene, as needed	**NAMHG protocol** o Halothane ✓ Caffeine *Diagnosis: awake MH episode*	Normal histology and histochemistry Family history was not provided	[10]

o, negative; ✓, positive; MH, malignant hyperthermia; URI, upper respiratory infection; Ca^{2+}, calcium; IV, intravenous; Tc, core temperature; MHS, malignant hyperthermia susceptible; MHE, malignant hyperthermia equivocal; MHN, malignant hyperthermia negative; IVCT, in vitro contracture test; EHS, exertional heat stroke; CPK, creatine phosphokinase; NAMHG, North American Malignant Hyperthermia Group; EMHG, European Malignant Hyperthermia Group; RyR1, ryanodine receptor 1; ER, exertional rhabdomyolysis; SkM, skeletal muscle

and include calcium voltage-gated channel subunit alpha1 S (CACNA1S), SH3 and cysteine rich domain 3 (STAC3) and calsequestrin 1 (CASQ1), all of which are tightly involved in Ca^{2+} homeostasis in skeletal muscles [42–45]. On the other hand, the effectiveness of DS for the prevention of EHS has been tested [46] with equivocal mixed results that require further validation [47–50]. This topic will be discussed in more detail later in this review and represents the second piece of evidence that EHS and MH may not share a mechanistic link.

4 Malignant Hyperthermia Diagnostics and Implications for Exertional Heat Stroke

Early recognition of an MH crisis is essential for the patient's survival. The clinical signs associated with MH are not unique and are almost non-specific. Therefore, anesthesiologists must be able to recognize a pattern of signs to make a rapid diagnosis [26]. Besides early clinical recognition of signs and symptoms, patients suspected of having MH usually undergo alternative protocols for diagnosis. To diagnose MH, the in vitro contracture test (IVCT) was developed and standardized 50 years ago [51–53] and remains the gold-standard. The test consists of a skeletal muscle biopsy from the *vastus lateralis* or the *biceps brachii* [9, 54]. To ensure viability, the sample is immediately mounted in an oxygenated organ bath at a standardized length and tension. The muscle sample is exposed to halothane or caffeine and fluctuations in muscle tension are recorded [55]. Halothane is a haloalkane anesthetic that acts upon the RyR1 channel [56]. Caffeine has a similar allosteric effect upon binding to the RyR1 channel to induce Ca^{2+} release from the sarcoplasmic reticulum [57].

Two governing bodies, the European Malignant Hyperthermia Group (EMHG) and the North American Malignant Hyperthermia Group (NAMHG) have characterized MH patients into three risk groups: 'malignant hyperthermia susceptible' (MHS), 'malignant hyperthermia negative' (MHN), and 'malignant hyperthermia equivocal' (MHE) [58]. These governing bodies have different ways to perform the IVCT protocol, which affects the overall sensitivity and specificity of the results [59, 60].

The EMHG uses two examinations known as the independent static cumulative halothane test and the independent static cumulative caffeine test [55]. The halothane test utilizes four incremental stages of concentration (0.5, 1.0, 2.0, and 3.0%) with each stage lasting 3 min or until a clinically relevant increase in tension is observed. A positive MH test is defined as a sustained increase of 0.2 g of baseline tension at a concentration of 2.0% or less. The caffeine challenge includes seven stages of concentrations (0.25, 0.5, 1.0, 2.0, 3.0, and 4.0 mmol/l) with the addition of positive control concentrations ranging from 8.0 to greater than 32.0 mmol/l [53, 54, 61, 62]. Each stage lasts 3 min and a positive MH test is considered when a sustained increase of at least 0.2 g of baseline tension at a concentration of 1.5 mmol or less is observed.

The NAMHG version of the protocol uses both caffeine and halothane differently. The test is referred to as the caffeine halothane contracture test (CHCT). The caffeine challenge is very similar to the EMHG protocol, but a positive MH diagnosis is considered if > 0.3 g of baseline tension is observed in response to 2.0 mmol/l or lower. The most important difference is within the halothane test. Only a single stage of 3% halothane exposure for 10 min is used and an increase > 0.5 g of baseline tension results in an MH positive diagnosis [60, 63]. Despite different protocols, the EMHG and NAMHG have adopted similar criteria for risk classification. That is, an increase in muscle tension in response to both halothane and caffeine stimulation results in a positive MHS diagnosis, while no changes or decreases in tension in response to both drugs results in a negative MH diagnosis (Table 2). However, EMHG recognizes either halothane or caffeine-only positive responses as MHE, whereas NAMHG recognizes a halothane-only positive response as MHE, but not caffeine-only [64].

A major limitation of these protocols is the lack of specificity for the MH condition. Numerous patients with neuromuscular diseases or previous episodes of heat-related illness have tested positive to these protocols [19, 62]. For instance, Heytens et al. [62] investigated a variety of neuromuscular diseases within the parameters of the EMHG protocol. Of the 60 patients, 2 were diagnosed MHS and another 10 diagnosed MHE. Those diseases testing positive ranged from Becker's muscular dystrophy and exertional myalgia to patients with inflammatory myopathies, chronic increased creatine kinase (CK), and/or skeletal muscle weakness [62]. Therefore, it is recommended to screen for neuromuscular diseases, such as Duchenne muscular dystrophy or central core disease as an exclusion criterion [65]. A previous episode of heat-related illness, which includes EHS, seems to influence the test outcome. For instance, Sagui et al. [19] employed the IVCT with biopsied *vastus lateralis* muscle from subjects who had previous episodes of EHS. Of the 454 subjects, 45.6% tested positive. Perhaps this represents the one and only piece of correlative, rather than causative, evidence available to suggest that EHS could be associated with or can cause Ca^{2+} dysregulation in individuals with no prior events of MH. While it is intriguing that previous EHS victims responded positively to the IVCT, these results must be interpreted with caution since the test has never been validated for EHS diagnosis, as Ca^{2+} dysregulation is not yet a known mechanism of EHS pathophysiology.

Table 2 In vitro contracture test (IVCT) protocols employed by the European Malignant Hyperthermia Group (EMHG) and the North American Malignant Hyperthermia Group (NAMHG)

Test	Halothane Protocol	Caffeine Protocol	MHS	MHE	MHN	Sensitivity (%)	Specificity (%)
EMHG	0.5, 1.0, 2.0, 3.0%; 3 min each stage, incremental increase [51, 52] **Positive test** = 0.2 g tension at 2.0% halothane or lower [52]	0.25, 0.5, 1.0, 1.5, 2.0, 3.0, 4.0, 32.0 mM; 3 min each stage, incremental increase [51, 52] **Positive test** = 0.2 g tension at 1.5 mM caffeine or lower	Both tests are positive	One of the tests is positive	Neither test is positive	99	93.6
NAMHG	One stage of 3% for 10 min **Positive test** = 0.5 g or higher tension [59]	0.5, 1.0, 2.0, 4.0, 8.0, 32.0 mM; 4 min each stage, incremental increase **Positive test** = 0.3 g or higher tension at 2.0 mM caffeine or lower		Positive only to halothane		97	78

MHS, malignant hyperthermia susceptible; MHE, malignant hyperthermia equivocal; MHN, malignant hyperthermia negative; mM, milimolar; g, grams; %, percentage

Misdiagnosis could lead to improper treatment or the sacrifice of one's career for the military or other occupational specialties [66]. Furthermore, despite the ability to classify patients' risk to develop a MH episode with these IVCT protocols, the NAMHG declared that the test cannot be solely used for the diagnosis of MH events or susceptibility, but may be a useful research tool to understand this condition [64]. The clinical signs should be evaluated in the context of the environmental factors that precipitated the event, family history and available medical data at the time of presentation for proper assessment and treatment [64]. Therefore, due to the lack of specificity, there is a need for improvement or replacement of the IVCT with an alternative protocol that demonstrates higher specificity. Due to the lack of specificity, many clinicians have increasingly implemented genetic testing to verify IVCT results and MH diagnosis in patients. These genetic tests normally target RYR1, CACNA1S, STAC3 and CASQ1 genes for already observed or novel single nucleotide polymorphisms that are associated with MH [44, 45, 67].

5 Similarities Between "Awake" Malignant Hyperthermia and Exertional Heat Stroke

Exertion in hot and humid environments has been shown to elicit both EHS and 'awake' MH episodes [8, 11, 68–70]. Awake cases of MH are similar in manifestation to EHS and may be a confounding factor when determining overlapping mechanisms between EHS and MH [68]. Awake MH is relatively rare, in comparison to anesthesia-induced MH, and accompanied by aggressive skeletal muscle cramping and rigidity, similar in manifestation to its anesthetic counterpart, but occurs in response to emotional stress, fatigue, strenuous exercise, heat exposure, or other unidentified causes [11, 68] (Table 1). Several risk factors are thought to predispose to an awake MH episode including family history of MH related to RyR1 mutations, pre-existing muscular conditions, recent or concurrent illness, or prior stress (psychological or environmental; Table 1) [8, 11, 68].

Awake MH has been reported in pig MH models [71] and, more rarely, in human case reports [68]. Autosomal dominant inheritance of RyR1 mutations makes a family history of MH the strongest documented risk factor for susceptibility (Table 1). A brief review reported cases from a seven-generation family that showed high incidence of vague lower limb pain, muscle cramps, poor exercise tolerance in hot weather, and stress-induced fatal MH episodes (in the absence of anesthesia exposure) [7]. A family with known MH mutations experienced such a high risk of MH fatalities associated with car transport (1 of every 60 relatives) that emotional stress was recognized as a triggering factor in humans, similar to the porcine condition [72]. It is now accepted that the porcine stress syndrome is the same genetic condition as MH due to stress intolerance, muscle rigor, cardiac arrhythmias, and development of exertional rhabdomyolysis during emotionally stressful conditions [7, 11]. However, the higher incidence of awake MH in pigs as opposed to the relatively low incidence in humans is due to presence of the RyR1 mutation in almost all porcine breeds [71]. Attempts to cool awake MH patients with ice packs, cool IV fluids or non-steroidal anti-inflammatory drugs (e.g.,

acetaminophen, aspirin) have proven ineffective, whereas DS is often prescribed prophylactically to these patients, thus highlighting the positive Ca^{2+} feedback loop in skeletal muscles as the mechanism for awake MH [8, 11, 13, 14, 68, 73].

Patients with a large muscular build (whether inherited or recently acquired) appear to have a predisposition towards recurrent muscle cramps, "fever" of unknown origin, and exertional rhabdomyolysis that can culminate in a full-blown awake MH episode [10–12, 73] (Table 1). In many case reports, there was a precipitating event (prior anesthesia exposure, illness, trauma, drug use) that may be linked to the MH episode. In other cases, skeletal muscle myopathies, such as central core disease with RyR1 mutations, have provided a mechanistic explanation for the incidence of awake MH [15, 73].

6 Treatment of Malignant Hyperthermia and Exertional Heat Stroke

There is no doubt that DS constitutes the most effective method to counteract both anesthesia-induced and awake MH events [11]. Nevertheless, whether it can be used as an effective treatment for EHS victims remains an intriguing question [46]. DS is a fast-acting postsynaptic muscle relaxant that lessens excitation–contraction coupling in muscle cells. It inhibits Ca^{2+} release from sarcoplasmic reticulum stores by antagonizing ryanodine receptors [74, 75]. DS has become an operating room staple since the discovery of anesthetically induced MH. Even though most of the haloalkane anesthetics known to trigger MH are no longer used, DS remains due to its effectiveness to suppress the release of Ca^{2+}, the element required by skeletal muscle to produce normal mechanical force and the unregulated, lethal catalyst of MH. Since MH is a mutative disease involving the RyR1 or other Ca^{2+} regulatory channels [5], DS is the gold-standard of treatment upon recognition of both anesthesia-induced and awake MH symptoms [75].

While several studies have attempted to determine the efficacy of DS for the treatment of EHS, conflicting results and reports of skeletal muscle weakness, loss of coordination and cardiac arrhythmias as negative side effects of DS raise concern regarding its potential use for the treatment of EHS [11, 46]. Routinely, after administration of DS, the patient would describe "weak legs, a 'funny' sensation in his muscles, and loss of coordination." Despite these potential side effects, the ability of DS to inhibit Ca^{2+} release has provided a rationale to test the efficacy of this intervention in EHS victims and animal models. Some authors report DS as being a more effective treatment for EHS than passive cooling, although this is not a universal finding [46–50]. Nevertheless, since an EHS episode, in a victim without any MH complications, is not induced by a pharmacokinetic imbalance, and victims treated in close proximity to the onset of the EHS routinely react well to submersion in cold ice water or ice packs, it does not seem appropriate to administer DS, especially in the face of the potential for the side effects that overlap with the normal response to EHS. These may include cardiac arrhythmias, hyperkalemia or tissue necrosis, although this is not an exhaustive list. Therefore, the risks of administering DS in all EHS victims outweigh the potential unproven benefits. Pre-clinical studies exploring the pharmacological potential of DS as a treatment for EHS are warranted.

7 Conclusions

Awake MH and EHS are associated with similar clinical symptomology. Therefore, they are proposed to be mechanistically linked to Ca^{2+} dysregulation in skeletal muscle (primarily through the RyR1; [16, 17, 19]). However, there is a lack of robust evidence to indicate a mechanistic link between the conditions for several reasons discussed in this review. First, the key distinguishing feature of MH is skeletal muscle or jaw rigidity and rapid development of hyperthermia that is resistant to external cooling with ice packs or similar interventions. Conversely, EHS patients present with skeletal muscle flaccidity and the most effective treatment has been firmly established as rapid external cooling [24]. Other pharmacological approaches (e.g. non-steroidal anti-inflammatory drugs) are not efficacious for either condition, but DS is highly effective in MH patients when provided early in the course of the syndrome. Importantly, DS has been tested for efficacy in EHS and shown either a lack of clinical efficacy or mixed results [46–50]. Second, there are no data, besides IVCT in samples taken from subjects who previously experienced heat-related illnesses, to indicate that EHS occurs due to Ca^{2+} dysregulation in the skeletal muscle except in those patients that have presented with muscle rigidity and were responsive to DS—and perhaps should be diagnosed as awake MH victims. This is a significant knowledge gap that could be addressed with pre-clinical studies to determine efficacy as well as potential side effects. Until these studies are conducted, the role of Ca^{2+} dysregulation in EHS will remain conjecture with no substantial supporting evidence. Third, awake MH cases occur predominantly in individuals that were previously exposed to anesthesia, heat, extreme or unaccustomed exercise or a combination of these factors [8, 11, 68]. While this is similar to EHS, there is no evidence that previous exposure to anesthesia increases susceptibility to EHS per se. Given the sensitivity of the RyR1 mutation to anesthesia, one would surmise prior anesthesia exposure to also increase the risk of EHS. From a clinical perspective, some aspects remain open for debate

and experimentation. For instance, the proper management of anesthesia in patients who have had a previous history of EHS remains a matter of debate. Additionally, there is not a clear statement as to whether individuals with a history of MH should be discouraged from participating in strenuous exercise in hot and humid environments. Again these represent areas of experimental investigation that could be explored to close these knowledge gaps to clearly identify a causal relationship between Ca^{2+} dysregulation and EHS. Although it is a provocative hypothesis to suggest that the similar clinical symptomology of MH and EHS may be indicative of a causal link between the conditions that would warrant genetic testing or the use of DS for both conditions, additional research is needed to provide a link that would warrant these measures.

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Compliance with Ethical Standards

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